

Worksheet W8: The Fundamental Theorem

"The Great Reversal — Differentiation and integration are opposites."

Chapter 8 · 12 Problems · Name: _____ Date: _____

Fundamental Theorem: The derivative and integral undo each other.

Problem 1

Question: What does the **Fundamental Theorem of Calculus** say?

💡 *Hint: Think about addition and subtraction — they undo each other.*

Problem 2

Question: What is the **antiderivative** of $2x$?

💡 *Hint: What function has derivative $2x$? Don't forget the constant C .*

Problem 3

Question: What is the **antiderivative** of $3x^2$?

💡 *Hint: What function has derivative $3x^2$? Don't forget C .*

Problem 4

Question: What is the **antiderivative** of x^3 ?

💡 *Hint: Add 1 to the exponent, then divide by the new exponent.*

Problem 5

Find the area under $y = x^2$ from $x = 0$ to $x = 4$ using the Fundamental Theorem.

💡 Hint: Antiderivative of x^2 is $x^3/3$. Compute $F(4) - F(0)$.

Problem 6

Question: What is the **constant of integration** and why do we add it?

💡 Hint: What happens to constants when you take the derivative?

Problem 7

Find the area under $y = 2x$ from $x = 0$ to $x = 5$.

💡 Hint: Antiderivative of $2x$ is x^2 . Compute $F(5) - F(0)$.

Problem 8

Find the area under $y = x^2 + 1$ from $x = 0$ to $x = 3$.

💡 Hint: Antiderivative of $x^2 + 1$ is $x^3/3 + x$.

Problem 9

Question: If you start with $f(x) = x^3$, take its derivative, and then integrate, what do you get?

💡 Hint: This is the "round trip" — function $\rightarrow$ derivative $\rightarrow$ integral.

Problem 10

Use the Fundamental Theorem to find $\int_1^4 3x^2 \, dx$.

💡 *Hint: Antiderivative of $3x^2$ is x^3 . Compute $F(4) - F(1)$.*

Problem 11

Question: Complete the "round trip" diagram: $f(x) = x^2 \xrightarrow{\text{Derivative}} \text{______} \xrightarrow{\text{Integral}} \text{______}$

💡 *Hint: Derivative of x^2 is $2x$. Integral of $2x$ is $x^2 + C$.*

Problem 12

Question: If the derivative and integral are opposites, why is the Fundamental Theorem called the "Great Reversal"?

💡 *Hint: It reversed how people thought about slope and area.*